Monitorovanie systému

UBUNTU SERVER

Monitorovanie

Monitorovanie jednotlivých serverov a služieb je dôležitou súčasťou administrácie systému.

Väčšina sieťových služieb sa monitoruje kvôli zisteniu výkonu, zisteniu dostupnosti, alebo kvôli zisteniu oboch týchto parametrov.

Na takéto monitorovanie sa používa systém Nagios, ktorý monitoruje dostupnosť a systém Munin, ktorý monitoruje výkon.

Taktiež sú rôzne ďalšie monitorovacie systémy, ako Webmin a podobne.

Monitorovanie - Nagios

Inštalácia

- sudo apt update

- Nutne balíky pre inštaláciu
  - sudo apt install -y autoconf bc gawk dc build-essential gcc libc6 make wget unzip apache2 php libapache2-mod-php libgd-dev libmcrypt-dev make libssl-dev snmp libnet-snmp-perl gettext
- Je potrebne si stiahnuť aktuálnu verziu Nagios Core 4.5.9
  - Manuálne na stránke
  - alebo
  - cd ~/
  - wget https://github.com/NagiosEnterprises/nagioscore/archive/nagios-4.5.9.tar.gz
- Rozbalíme stihnutý súbor:
  - tar -xf nagios-4.5.9.tar.gz
  - cd nagioscore-*/

Monitorovanie - Nagios

Je potrebne urobiť kompiláciu:

- sudo ./configure --with-httpd-conf=/etc/apache2/sites-enabled
- sudo make all

```
*** Configuration summary for nagios 4.4.6 2020-04-28 ***:

 General Options:
 -------------------------
        Nagios executable:  nagios
        Nagios user/group:  nagios,nagios
       Command user/group:  nagios,nagios
             Event Broker:  yes
        Install ${prefix}:  /usr/local/nagios
    Install ${includedir}:  /usr/local/nagios/include/nagios
                Lock file:  /run/nagios.lock
   Check result directory:  /usr/local/nagios/var/spool/checkresults
           Init directory:  /lib/systemd/system
  Apache conf.d directory:  /etc/apache2/sites-enabled
             Mail program:  /bin/mail
                  Host OS:  linux-gnu
          IOBroker Method:  epoll

 Web Interface Options:
 ------------------------
                 HTML URL:  http://localhost/nagios/
                  CGI URL:  http://localhost/nagios/cgi-bin/
 Traceroute (used by WAP):


Review the options above for accuracy.  If they look okay,
type 'make all' to compile the main program and CGIs.

root@nagios20 ~/nagioscore-nagios-4.4.6# sudo make all
```

Monitorovanie - Nagios

Vytvorte používateľa a skupinu Nagios:

- sudo make install-groups-users
- sudo usermod -a -G nagios www-data

```
root@nagios20 ~/nagioscore-nagios-4.4.6#
root@nagios20 ~/nagioscore-nagios-4.4.6#
root@nagios20 ~/nagioscore-nagios-4.4.6# sudo make install-groups-users
groupadd -r nagios
useradd -g nagios nagios
root@nagios20 ~/nagioscore-nagios-4.4.6# sudo usermod -a -G nagios www-data
root@nagios20 ~/nagioscore-nagios-4.4.6#
```

Nainštalujte binárne súbory Nagios:

- sudo make install
- sudo make install-daemoninit
- sudo make install-commandmode

Monitorovanie - Nagios

Potom nainštalujte skompilovanú konfiguráciu pomocou skriptu:

sudo make install-config

Potom nainštalujte konfiguráciu Apache pre Nagios a aktivujte moduly mod_rewrite a mode_cgi.

- sudo make install-webconf
- sudo a2enmod rewrite cgi

Reštartujte službu Apache.

- systemctl restart apache2

Monitorovanie - Nagios

```
*** Config files installed ***

Remember, these are *SAMPLE* config files.  You'll need to read
the documentation for more information on how to actually define
services, hosts, etc. to fit your particular needs.

root@nagios20 ~/nagioscore-nagios-4.4.6# sudo make install-webconf
/usr/bin/install -c -m 644 sample-config/httpd.conf /etc/apache2/sites-enabled/nagios.conf
if [ 0 -eq 1 ]; then \
        ln -s /etc/apache2/sites-enabled/nagios.conf /etc/apache2/sites-enabled/nagios.conf; \
fi

*** Nagios/Apache conf file installed ***

root@nagios20 ~/nagioscore-nagios-4.4.6# sudo a2enmod rewrite cgi
Enabling module rewrite.
Enabling module cgi.
To activate the new configuration, you need to run:
  systemctl restart apache2
root@nagios20 ~/nagioscore-nagios-4.4.6# systemctl restart apache2
root@nagios20 ~/nagioscore-nagios-4.4.6#
```

- Vytvorte nového Apache používateľa „nagiosadmin“.
  - sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin

Monitorovanie - Nagios

```
root@nagios20 ~/nagioscore-nagios-4.4.6#
root@nagios20 ~/nagioscore-nagios-4.4.6#
root@nagios20 ~/nagioscore-nagios-4.4.6# sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
root@nagios20 ~/nagioscore-nagios-4.4.6#
root@nagios20 ~/nagioscore-nagios-4.4.6# cat /usr/local/nagios/etc/htpasswd.users
nagiosadmin:$apr1$Mksj.r.E$aJEIPUGco8F8krUBJf1//1
root@nagios20 ~/nagioscore-nagios-4.4.6#
root@nagios20 ~/nagioscore-nagios-4.4.6#
```

- Nastavenie Firewall
  - Pre konfiguráciu firewallu potrebne povolit službu Apache a port pre server Nagios do firewallu UFW.
    - for svc in Apache ssh
    - do
    - ufw allow $svc
    - done

Monitorovanie - Nagios

Zapneme UFW:

- sudo ufw enable

Teraz skontrolujte všetky dostupné pravidlá pomocou príkazu nižšie.

- ufw status numbered

```
root@nagios20 ~/nagioscore-nagios-4.4.6#
root@nagios20 ~/nagioscore-nagios-4.4.6#
root@nagios20 ~/nagioscore-nagios-4.4.6# for svc in Apache ssh
do
ufw allow $svc
done
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
root@nagios20 ~/nagioscore-nagios-4.4.6# ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@nagios20 ~/nagioscore-nagios-4.4.6#
root@nagios20 ~/nagioscore-nagios-4.4.6# ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] Apache                     ALLOW IN    Anywhere
[ 2] 22/tcp                     ALLOW IN    Anywhere
[ 3] Apache (v6)                ALLOW IN    Anywhere (v6)
[ 4] 22/tcp (v6)                ALLOW IN    Anywhere (v6)

root@nagios20 ~/nagioscore-nagios-4.4.6#
```

Monitorovanie - Nagios

Nainštalujeme Nagios Pluginy a NRPE Plugin

- sudo apt install monitoring-plugins nagios-nrpe-plugin

Po dokončení inštalácie prejdite do inštalačného adresára nagios "/usr/local/nagios/etc" a vytvorte nový adresár na uloženie konfigurácie pre všetkých používateľov servera.

- cd /usr/local/nagios/etc
- mkdir -p /usr/local/nagios/etc/servers

Potom upravte konfiguráciu Nagios 'nagios.cfg' pomocou editora nano.

- sudo nano nagios.cfg

Odkomentujte možnosť 'cfg dir', ktorá sa použije na uloženie všetkých konfigurácií serverov.

- cfg_dir=/usr/local/nagios/etc/servers

Monitorovanie - Nagios

Potom upravte konfiguračný súbor "resource.cfg" a definujte cestu k binárnym súborom Nagios Monitoring Plugins.

- sudo nano resource.cfg

Definujte cestu k modulom Nagios Monitoring Plugins zmenou predvolenej konfigurácie.

$USER1$=/usr/lib/nagios/plugins

Potom pridajte e-mailový kontakt pre správcu nagios úpravou konfiguračného súboru „objects/contacts.cfg“.

- sudo nano objects/contacts.cfg

```
define contact{
        ......
        email          email@host.com
}
```

Monitorovanie - Nagios

Teraz definujte pridanie nrpe check skriptu do konfiguračného súboru "objects/commands.cfg".

- sudo nano objects/commands.cfg

```
define command{
    command_name check_nrpe
    command_line $USER1$/check_nrpe -H $HOSTADDRESS$ -c $ARG1$
}
```

Ďalej spustite službu Nagios a pridajte ju do bootovania systému.

- sudo systemctl start nagios
- sudo systemctl enable nagios

Monitorovanie - Nagios

```
root@nagios20 ~#
root@nagios20 ~# cd /usr/local/nagios/etc
root@nagios20 /u/l/n/etc#
root@nagios20 /u/l/n/etc# mkdir -p /usr/local/nagios/etc/servers
root@nagios20 /u/l/n/etc#
root@nagios20 /u/l/n/etc# ls
cgi.cfg  htpasswd.users  nagios.cfg  objects  resource.cfg  servers
root@nagios20 /u/l/n/etc#
root@nagios20 /u/l/n/etc# vim nagios.cfg
root@nagios20 /u/l/n/etc#
root@nagios20 /u/l/n/etc# vim resource.cfg
root@nagios20 /u/l/n/etc#
root@nagios20 /u/l/n/etc# vim objects/contacts.cfg
root@nagios20 /u/l/n/etc#
root@nagios20 /u/l/n/etc# vim objects/commands.cfg
root@nagios20 /u/l/n/etc#
root@nagios20 /u/l/n/etc# systemctl start nagios
root@nagios20 /u/l/n/etc# systemctl enable nagios
root@nagios20 /u/l/n/etc#
root@nagios20 /u/l/n/etc#
```

Skontrolujeme stav systemu:

- sudo systemctl status nagios

Monitorovanie - Nagios

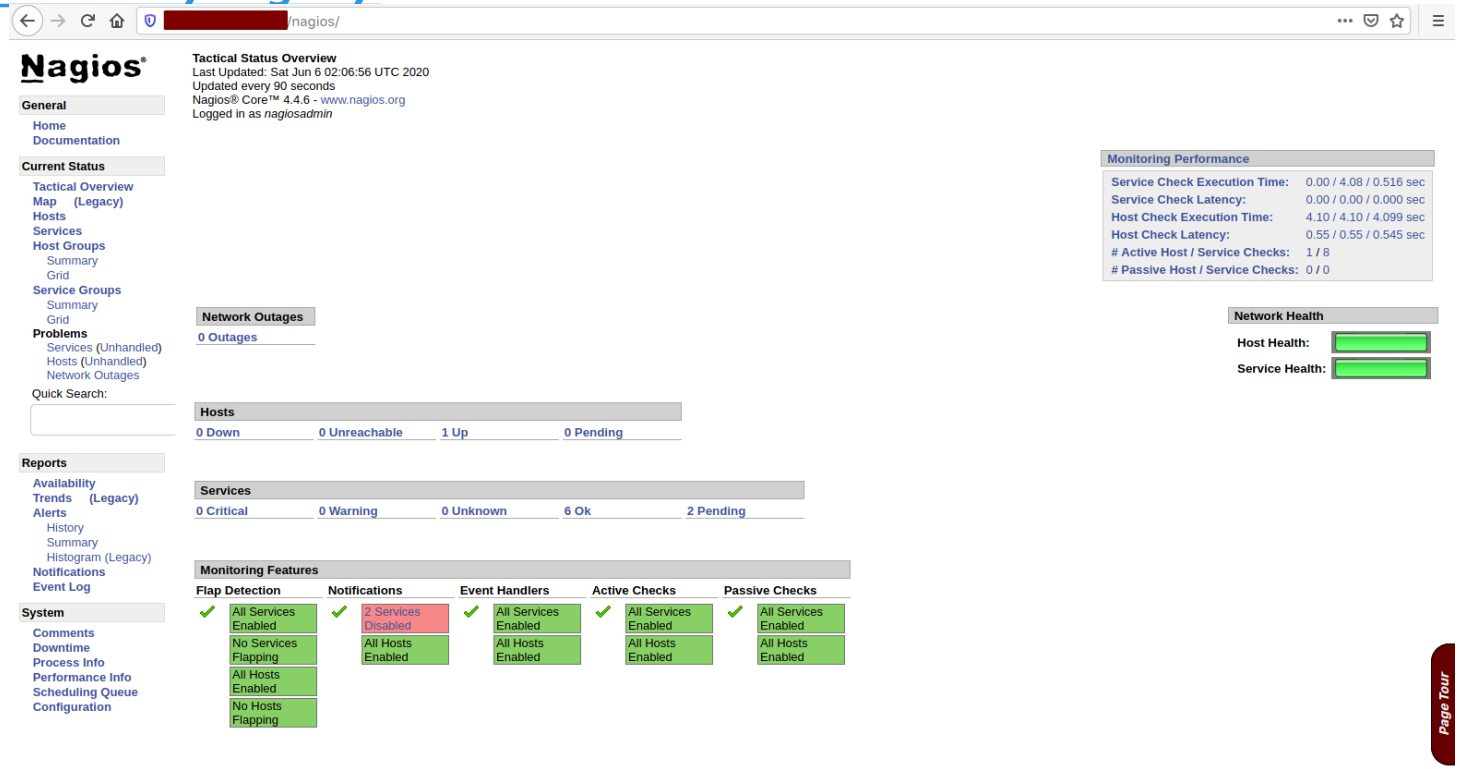
```
root@nagios20 /u/l/n/etc#
root@nagios20 /u/l/n/etc# systemctl restart apache2
root@nagios20 /u/l/n/etc# systemctl status nagios
● nagios.service - Nagios Core 4.4.6
     Loaded: loaded (/lib/systemd/system/nagios.service; enabled; vendor preset: enabled)
     Active: active (running) since Sat 2020-06-06 02:02:40 UTC; 1min 11s ago
       Docs: https://www.nagios.org/documentation
   Main PID: 44660 (nagios)
      Tasks: 6 (limit: 1074)
     Memory: 2.5M
     CGroup: /system.slice/nagios.service
             ├─44660 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             ├─44665 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─44666 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─44667 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             ├─44668 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
             └─44728 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
```

Následne je potrebne reštartovať Apache server, aby sa aplikovali zmeny:

- sudo systemctl restart apache2

Monitorovanie - Nagios

http://xxx.xxx.xxx.xxx/nagios/



Monitorovanie - Munin

Inštalácia:

- sudo apt update

Potrebujeme nakonfigurovať webový server na Ubuntu na prístup k webovému rozhraniu cez sieť Munin. Na inštaláciu Apache a ďalších požadovaných balíkov preto použite príkaz uvedený nižšie. V prípade že je Apache už nainštalovaný tento príkaz vynecháme

- sudo apt install apache2 libcgi-fast-perl libapache2-mod-fcgid
- sudo apt install munin

Ak server, ktorý chceme monitorovať, má aj databázu Mysql alebo MariaDB, potom na jeho monitorovanie potrebujeme aj nasledujúce doplnky Munin.

Monitorovanie - Munin

```
cd /etc/munin/plugins
ln -s /usr/share/munin/plugins/mysql_ mysql_
ln -s /usr/share/munin/plugins/mysql_bytes mysql_bytes
ln -s /usr/share/munin/plugins/mysql_innodb mysql_innodb
ln -s /usr/share/munin/plugins/mysql_isam_space_ mysql_isam_space_
ln -s /usr/share/munin/plugins/mysql_queries mysql_queries
ln -s /usr/share/munin/plugins/mysql_slowqueries mysql_slowqueries
ln -s /usr/share/munin/plugins/mysql_threads mysql_threads
```

Taktiež je potrebne povoliť ďalšie protokoly a adresáre potrebné na ukladanie údajov generovaných Muninom.

- sudo nano /etc/munin/munin.conf

Monitorovanie - Munin

Odpoznámkujeme:

```
root@h2s-VirtualBox: /etc/munin/plugins
  GNU nano 4.8                  /etc/munin/munin.conf                  Modified
# Example configuration file for Munin, generated by 'make build'

# The next three variables specifies where the location of the RRD
# databases, the HTML output, logs and the lock/pid files.  They all
# must be writable by the user running munin-cron.  They are all
# defaulted to the values you see here.
#
dbdir   /var/lib/munin
htmldir /var/cache/munin/www
logdir /var/log/munin
rundir  /var/run/munin

# Where to look for the HTML templates
#
#tmpldir        /etc/munin/templates

# Where to look for the static www files
#
#staticdir /etc/munin/static

                              [ Read 145 lines ]
^G Get Help   ^O Write Out ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos
^X Exit       ^R Read File ^\ Replace    ^U Paste Text^T To Spell   ^_ Go To Line
```

Monitorovanie - Munin

Nakonfigurujte Apache pre monitorovanie servera Munin

Prejdite do konfiguračného adresára Apache:

- cd /etc/apache2/conf-enabled

Vytvorte symbolický odkaz

- ln -s /etc/munin/apache24.conf munin.conf

Následne editujeme tento súbor:

- sudo nano /etc/munin/apache24.conf

Monitorovanie - Munin

```
root@h2s-VirtualBox: /etc/apache2/conf-enabled
root@h2s-VirtualBox: /etc/apache2/c...    h2s@h2s-VirtualBox: ~
  GNU nano 4.8                    /etc/munin/apache24.conf
ScriptAlias /munin-cgi/munin-cgi-graph /usr/lib/munin/cgi/munin-cgi-graph
Alias /munin/static/ /var/cache/munin/www/static/

<Directory /var/cache/munin/www>
#    Require local
    Require all granted
 Options FollowSymLinks SymLinksIfOwnerMatch
</Directory>

<Directory /usr/lib/munin/cgi>
 #   Require local
Require all granted
 Options FollowSymLinks SymLinksIfOwnerMatch
    <IfModule mod_fcgid.c>
        SetHandler fcgid-script
    </IfModule>
    <IfModule !mod_fcgid.c>
        SetHandler cgi-script
    </IfModule>
</Directory>

^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos
^X Exit       ^R Read File  ^\ Replace    ^U Paste Text ^T To Spell   ^_ Go To Line
```

Monitorovanie - Munin

Reštartujte Apache:

- sudo systemctl restart apache2

Reštartujte službu Munin:

- sudo systemctl restart munin munin-node

Následné je potrebné skontrolovať stav:

- systemctl status munin munin-node

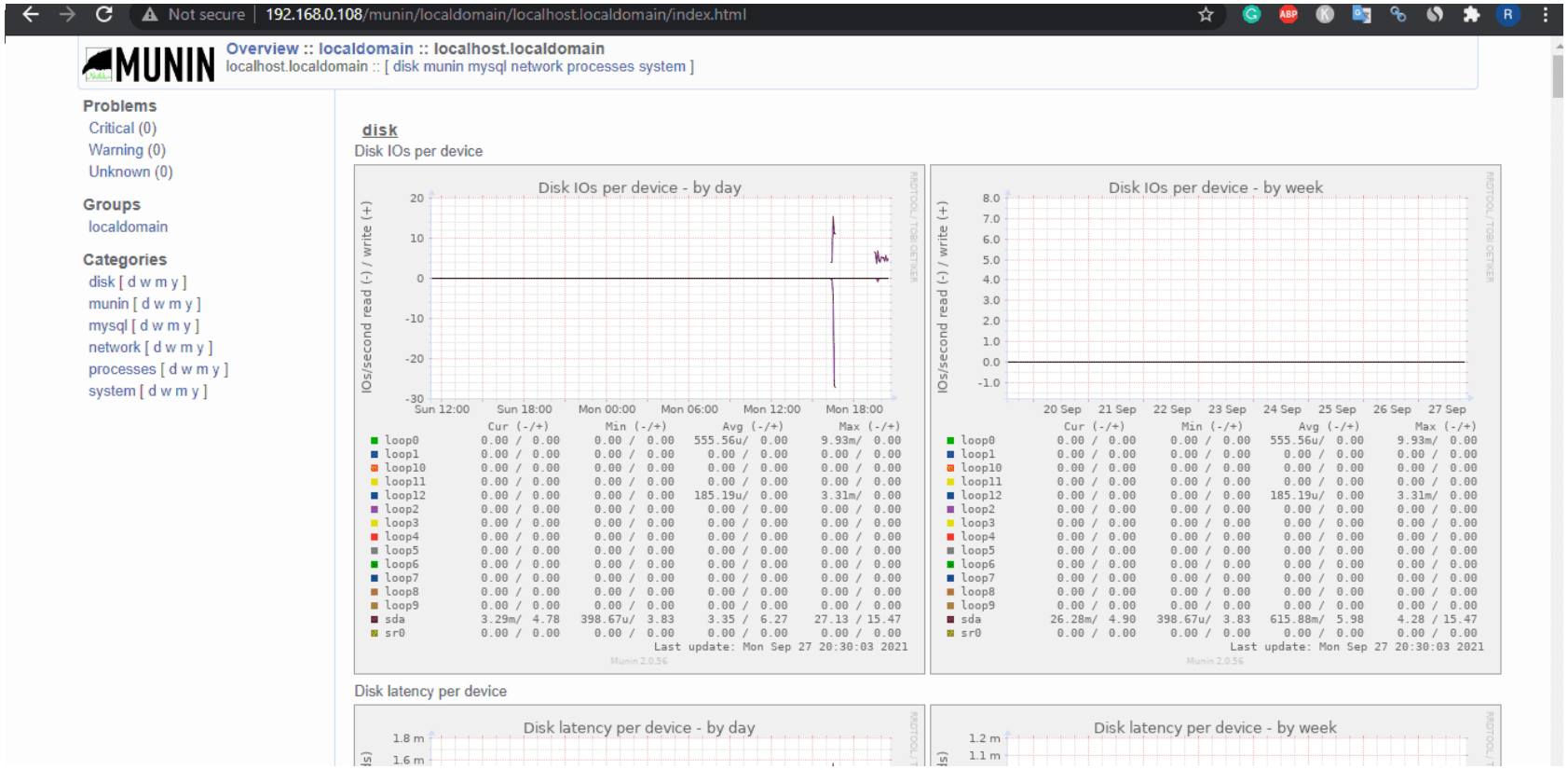
Monitorovanie - Munin

```
root@h2s-VirtualBox:/etc/apache2/conf-enabled# systemctl restart munin munin-nod
e
root@h2s-VirtualBox:/etc/apache2/conf-enabled# systemctl status munin munin-node
● munin.service - LSB: Create munin master directories on boot
     Loaded: loaded (/etc/init.d/munin; generated)
     Active: active (exited) since Mon 2021-09-27 20:25:54 IST; 15s ago
       Docs: man:systemd-sysv-generator(8)
    Process: 95402 ExecStart=/etc/init.d/munin start (code=exited, status=0/SUC>

Sep 27 20:25:54 h2s-VirtualBox systemd[1]: Starting LSB: Create munin master di>
Sep 27 20:25:54 h2s-VirtualBox systemd[1]: Started LSB: Create munin master dir>

● munin-node.service - Munin Node
     Loaded: loaded (/lib/systemd/system/munin-node.service; enabled; vendor pr>
     Active: active (running) since Mon 2021-09-27 20:25:54 IST; 15s ago
       Docs: man:munin-node(1)
             http://guide.munin-monitoring.org/en/stable-2.0/reference/munin-no>
    Process: 95405 ExecStart=/usr/sbin/munin-node $DAEMON_ARGS (code=exited, st>
   Main PID: 95406 (munin-node)
      Tasks: 1 (limit: 4651)
     Memory: 10.2M
     CGroup: /system.slice/munin-node.service
             └─95406 /usr/bin/perl -wT /usr/sbin/munin-node
```

Monitorovanie - Munin

http://server-ip-address/munin

https://p



Monitorovanie-Webmin

Inštalácia Webmin:

- sudo apt update

V prvom rade je potrebne vykonať editáciu súboru:

https://phoenixnap.com/kb/install-webmin-on-ubuntu

- sudo nano /etc/apt/sources.list

sudo nano /etc/apt/sources.list.d/webmin.list

Pridanie nového zdroja:

- *deb http://download.webmin.com/download/repository sarge contrib*

Potom si stiahnite kľúč Webmin PGP pomocou wget a pridajte ho do zoznamu kľúčov vášho systému:

- *wget -q -O- http://www.webmin.com/jcameron-key.asc | sudo apt-key add*

Monitorovanie-Webmin

Update:

- *sudo apt update*

Inštalácia Webmin:

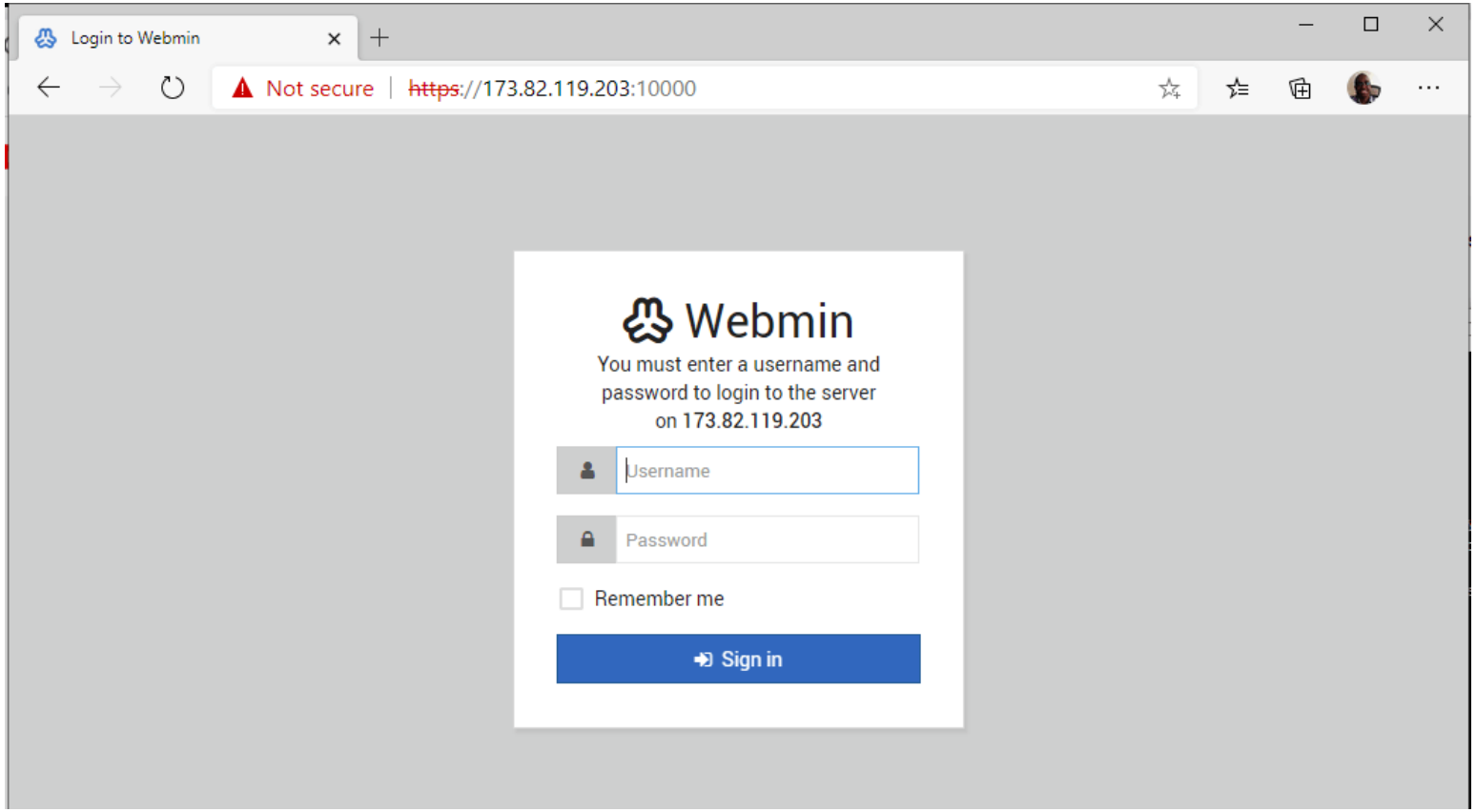
- *sudo apt install webmin*

```
Output
. . .
Webmin install complete. You can now login to
https://your_server:10000 as root with your
root password, or as any user who can use sudo.
```

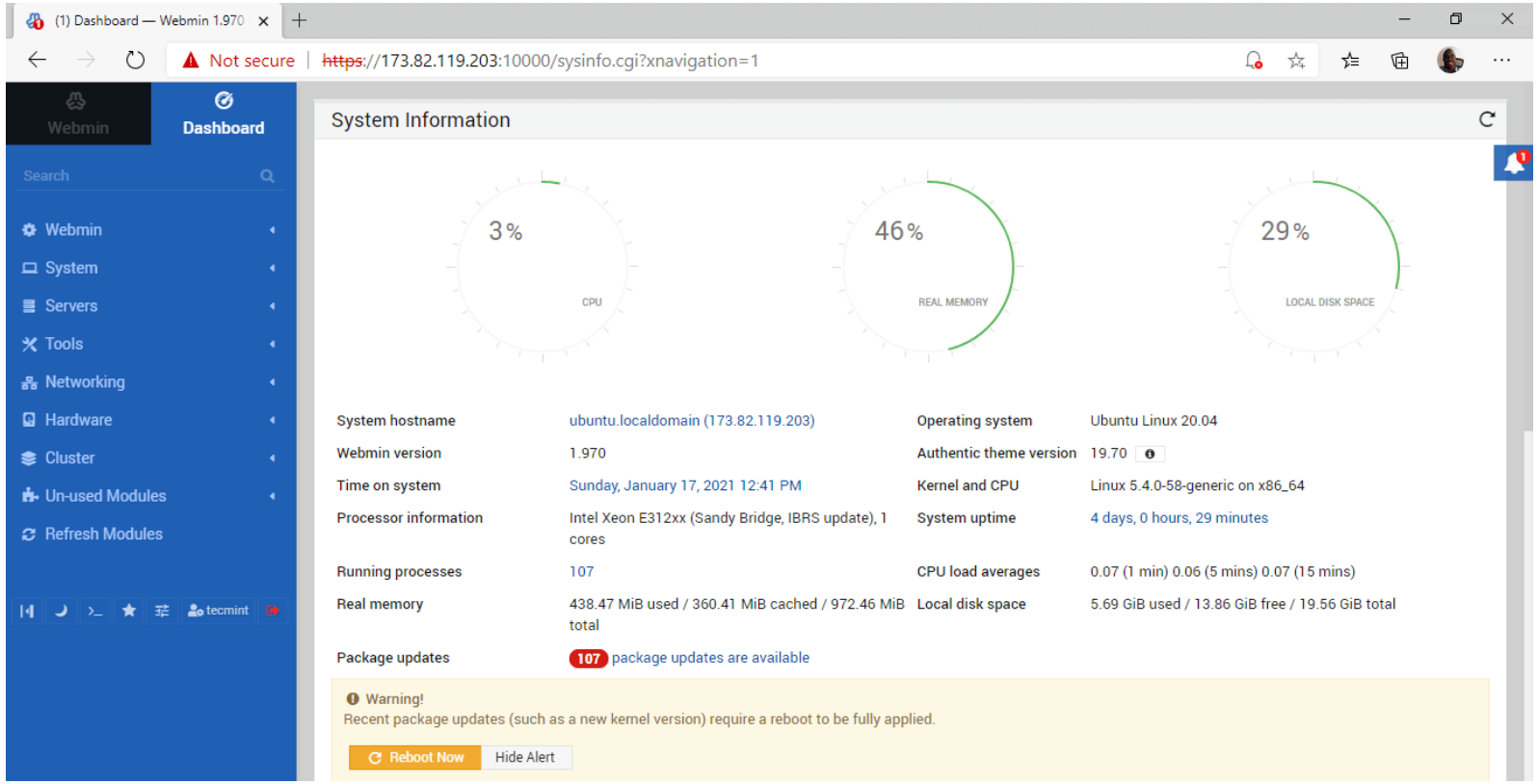
Ak sme povolili ufw, bude potrebné povoliť Webmin cez bránu firewall:

- *sudo ufw allow 10000*

Monitorovanie-Webmin



Monitorovanie-Webmin



Ďakujem za pozornosť